

1 Simple linear Regression

Reference: Chapters 1-2 of Draper and Smith.

1.1 Introduction

Set-up: There are n experimental units. On each unit we measure several variables. One particular variable is of interest (Y) : response. Other variables ($x_1, \dots, x_p$) are predictors.

Example1: unit=A day. (Y_i, x_{1i}, x_{2i}) observed for day $i = 1, \dots, n$

- Y = Total precipitation.
- X_1 = Temp at 10AM
- X_2 = Relative Humidity at 10AM.

Aim: We are asking for a function f that helps us predict the "best" value of Y given $(X_1, \dots, X_p)$.

1. This f is completely unknown.
2. **Simple:** $p = 1$. One can have $p > 1$. (Multiple Regression.)
3. **Linear:** f is a linear function, $f(x) = a + bx$. Drop Linearity: Non-linear Regression eg, $y_i = \alpha e^{\beta x_i} + \epsilon_i$, which is not a linear combination of the parameters.
4. For simple linear regression, Y is continuous. Y can be discrete/categorical data, logistic regression.

Why Regression?

1. To predict the value of Y given $X_1, \dots, X_p$.
2. To study the impact of a predictor on the response. $Y = a + bX$. What is the change in response for unit change in predictor? Estimator of Parameters.
3. Testing Hypothesis e.g. $\beta > 0$. Hypothesis about the nature of impact.

Example 2: unit= an individual. Y = Salary. x_1 = education, x_2 = gender, x_3 = experience

Keeping education and experience fixed, does gender affect Salary? (Testing).
How much does salary change for every year of experience? (Estimation)

1.2 Model

Data: $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

Model and Estimation:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad i = 1, 2, \dots, n$$

where β_0 and β_1 are unknown fixed parameters.

Assumptions

1. Linear Model holds
2. ϵ_i are independent of each other
3. ϵ_i have the same distribution for all values of x_i
4. Often x_i are fixed. Otherwise we condition on x_i .
5. $E[\epsilon_i] = 0$ $\text{Var}(\epsilon_i) = \sigma^2$ (some number)

ϵ_i are random, so Y_i are also random. $E[Y_i] = \beta_0 + \beta_1 x_i + 0$, $\text{Var}(Y_i) = \sigma^2$.

Parameters: $\beta_0, \beta_1, \sigma^2$.

Scatterplot First step is to draw a scatter plot and ensure linear model is reasonable at least visually. Plot of infant mortality (y) on female literacy (x). Roughly the pattern supports linearity assumption.

Goal:

1. Estimate Parameter $(\beta_0, \beta_1, \sigma^2)$ (fitting). This can be followed up with hypothesis tests and prediction of y given x .
2. Verify if model assumptions are satisfied. (Diagnostics)
If assumptions are not satisfied, we modify the model. We fit a model and get residuals from there.
3. Keep doing this until you get a model that satisfies all assumptions.

Least Squares Criterion

Next estimate the β 's & σ^2 . The goal is to find the "best" fitting line. What is "best"?

Given $y_1, \dots, y_n$ to predict one value of Y , the most common candidate is $\bar{y}$.

$$\arg \min_a \sum_{i=1}^n (Y_i - a)^2 = \bar{Y}$$

$\bar{y}$ minimizes the sum of squared distances from all obs.

Following this idea, in the regression set-up, we minimize the sum of squared distances between the observed and fitted values.

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2$$

Another possible criterion is **Least Absolute Deviation**: $\min \sum |y_i - \beta_0 - \beta_1 x_i|$.

Least Squares Estimator Solution:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Shift the line around to minimize the sum of squared vertical distance of the points in the scatterplot to the line.

Roles of predictor and response:

1. We do not consider perpendicular distances in regression.
2. Here we treat x & y asymmetrically and minimize the vertical distance of the points from the line.
3. If we regress x on y :

$$x = \alpha_0 + \alpha_1 y + \epsilon$$

is the model.

$$\hat{\alpha}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (y_i - \bar{y})^2}$$

$$\hat{\alpha}_0 = \bar{x} - \hat{\alpha}_1 \bar{y}$$

This minimizes horizontal distance.

4. Both fitted lines pass through $(\bar{x}, \bar{y})$

$$y - \bar{y} = \hat{\beta}_1 (x - \bar{x})$$

$$x - \bar{x} = \hat{\alpha}_1 (y - \bar{y})$$

$$\hat{\beta}_1 \hat{\alpha}_1 = r^2 \quad (\text{r is corr. coeff.})$$

Regression to the mean

The regression line can be written as:

$$y - \bar{y} = \hat{\beta}_1(x - \bar{x})$$

Standardizing x and y :

$$\frac{y - \bar{y}}{s_y} = r \cdot \frac{x - \bar{x}}{s_x}$$

where r is the correlation coefficient and $|r| \leq 1$.

For example, Galton observed that tall fathers have sons who are not as tall (on average, their height regresses towards the mean). Another example is scores of 2 tests of the same set of individuals. Those who perform very well in one test do not do that well, on the average, in the second test.

MLE: If ϵ_i are normally distributed, then the least squares estimator is the Maximum Likelihood Estimator (MLE). The least squares criterion is general and does not need a distributional assumption.

How to estimate σ^2 ?

The residuals are $\hat{\epsilon}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$. An unbiased estimator for σ^2 is:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{\epsilon}_i^2 = \frac{1}{n-2} \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

The factor $n-2$ is a correction factor to make the estimator unbiased. For a simple model $y_i = a + \epsilon_i$, the estimate for variance is $\frac{1}{n-1} \sum (y_i - \bar{y})^2$.

1.3 Model evaluation

Is x useful in predicting y ?

If there was no predictor, the error in predicting y would be measured by the **Total Sum of Squares (SST)**:

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

The **Residual Sum of Squares (SSR)** after fitting the model is:

$$SSRes = \sum (y_i - \hat{y}_i)^2 = \sum (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

If the regression exercise is useful, then $SSRes < SST$. The larger the difference ($SST - SSR$), the more useful (better) is the model.

Decomposition of Variance

Claim A: $SST \geq SSRes$.

Let $S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$, $S_{xx} = \sum (x_i - \bar{x})^2$, $S_{yy} = \sum (y_i - \bar{y})^2$. Recall $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ and $\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$.

$$\begin{aligned}
 SSRes &= \sum (y_i - (\bar{y} - \hat{\beta}_1 \bar{x}) - \hat{\beta}_1 x_i)^2 = \sum [(y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x})]^2 \\
 &= \sum (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum (x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2 \sum (x_i - \bar{x})^2 \\
 &= S_{yy} - 2\hat{\beta}_1 S_{xy} + \hat{\beta}_1^2 S_{xx} \\
 &= S_{yy} - 2 \frac{S_{xy}}{S_{xx}} S_{xy} + \left(\frac{S_{xy}}{S_{xx}} \right)^2 S_{xx} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}
 \end{aligned}$$

So, $SSR = SST - \frac{S_{xy}^2}{S_{xx}}$, which implies $SSR \leq SST$.

The amount by which the sum of squares is being reduced due to regression is the **Regression Sum of Squares (SSReg)**.

$$SSReg = SST - SSRes = \frac{S_{xy}^2}{S_{xx}}$$

Coefficient of Determination (R^2)

The ratio of SS explained by regression to the total SS of y is denoted by R^2 .

$$R^2 = \frac{SSReg}{SST} = \frac{S_{xy}^2 / S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx} S_{yy}} = r^2$$

where r is the correlation coefficient between x and y. R^2 is called the coefficient of determination. This is true in general, even with multiple predictors.

Analysis of Variance Table (ANOVA)

Source	SS	df	MS (Mean Square)	F
Regression	SSReg	1	$MSReg = \frac{SSReg}{1}$	$\frac{MSReg}{MSRes}$
Residual	SSRes	n-2	$MSRes = \frac{SSRes}{n-2}$	
Total	SST	n-1		

- Each SS has chi-square (χ^2) distribution with corresponding degrees of freedom (df).
- Larger values of R^2 are better.
- R^2 measures the proportion of variability of y that can be explained by linear regression on x.
- A larger F-statistic is associated with a larger R^2 .

- The ratio of two independent chi-square (χ^2) distributions (divided by their df) gives an F-distribution.
- So we can do a Hypothesis Test (under normality conditions).

Hypothesis Test for Significance of Regression

- H_0 : Regression is not significant ($\beta_1 = 0$).
- H_a : Regression is significant ($\beta_1 \neq 0$).

The F-statistic can be written in terms of R^2 :

$$F = \frac{MSReg}{MSRes} = \frac{SSReg/1}{SSRes/(n-2)} = \frac{R^2 \cdot SST}{(1-R^2)SST/(n-2)} = \frac{R^2}{1-R^2}(n-2)$$

If F is significant then the model has significantly reduced the variation in the data. This indicates that the model is useful, that is X is useful in predicting Y . Ideally we want a significant F statistic and a large R^2 . In most practical situations, F is significant but R^2 is moderate (around 0.5-0.6). This means that X is useful, but there is still a lot of unexplained variation in Y .

1.4 Diagnostics for Simple Linear Regression

1. **Linearity:** Plot residuals against x_i (or fitted values). If there's a pattern, the relationship may be non-linear.
2. **Independence of errors:** ϵ_i are independent of each other. This is difficult to check directly. Serial dependence is a particular for data from time series.
3. **Independence of errors and predictors:** ϵ_i are independent of x_i , x vs ϵ should have no pattern.
4. **Zero mean of errors:** $E(\epsilon_i) = 0$.
5. **Constant variance (Homoscedasticity):** $\text{Var}(\epsilon_i) = \sigma^2$. Plot residuals $\hat{\epsilon}_i$ against fitted values. A funnel shape suggests non-constant variance.
6. **Normality of errors:** If you assume ϵ_i are normal for tests, check this with a QQ-plot of residuals of $\hat{\epsilon}_i$.
7. **Leverage:** Identify points that have a high influence on the regression line. Leverage measures how much the regression line changes by the addition of a point.

1.5 Bivariate Normal & Relation to Simple Linear Regression

Reference for this section: Sec 2.11 of Kutner et al. Applied Linear Statistical Models

So far we have assumed that values of the x (the predictors) are fixed. That may be realistic in certain situations, e.g., Demand curve, that may be realistic. How demand changes due to change in Price (maybe double), However, in most cases, we can't really fix X .

e.g., the effect of age on BP, people are walking into a hospital clinic & you record their age & BP. In such situations, does the linear model still hold if we condition on X ?

The answer is yes, if $(X, Y) \sim N_2(\dots)$. The conditional distribution is given by:

$$Y|X = x \sim N(\beta_0 + \beta_1 x + \dots, \sigma^2)$$

This can be derived from the joint bivariate normal distribution. The conditional density is the ratio of the joint density to the marginal density:

$$f_{Y|X=x}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Let the joint distribution be:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2 \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \right)$$

Where $X \sim N(\mu_1, \sigma_1^2)$.

The joint PDF is:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp \left\{ \frac{-1}{2(1-\rho^2)} \left[\frac{(x-\mu_1)^2}{\sigma_1^2} - 2\frac{\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2} + \frac{(y-\mu_2)^2}{\sigma_2^2} \right] \right\}$$

The marginal PDF of X is:

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_1} \exp \left\{ -\frac{(x-\mu_1)^2}{2\sigma_1^2} \right\}$$

After dividing and simplifying the expression in the exponent, we get:

$$f_{Y|X=x}(y) = \frac{1}{\sqrt{2\pi}\sigma_2\sqrt{1-\rho^2}} \exp \left\{ \frac{-1}{2\sigma_2^2(1-\rho^2)} \left[y - \left(\mu_2 + \rho\frac{\sigma_2}{\sigma_1}(x-\mu_1) \right) \right]^2 \right\}$$

This shows that the conditional distribution of Y given $X = x$ is Normal:

$$Y|X = x \sim N\left(\mu_2 + \rho \frac{\sigma_2}{\sigma_1}(x - \mu_1), \sigma_2^2(1 - \rho^2)\right)$$

This is a simple linear regression model $Y = \beta_0 + \beta_1 x + \epsilon$, where.

$$\epsilon \sim N(0, \sigma_2^2(1 - \rho^2))$$

:

- Regression slope: $\beta_1 = \rho \frac{\sigma_2}{\sigma_1}$
- Regression intercept: $\beta_0 = \mu_2 - \beta_1 \mu_1$
- Error variance: $\sigma^2 = \text{Var}(\epsilon) = \sigma_2^2(1 - \rho^2)$

β_0 and β_1 are the population versions of the least squares estimates ($\hat{\beta}_{0,LS} = \bar{y} - \hat{\beta}_{1,LS}\bar{x}$ and $\hat{\beta}_{1,LS} = \frac{S_{xy}}{S_{xx}}$).

$$1 - R^2 = 1 - \frac{\text{SSReg}}{\text{SST}} = \frac{\text{SSRes}}{\text{SST}}$$

$$1 - \rho^2 = \frac{\sigma^2}{\sigma_2^2}, \quad \sigma^2 = \text{Var}(\epsilon), \quad \sigma_2^2 = \text{Var}(Y)$$

Summary When $(X, Y) \sim N_2$, the simple linear regression model holds. The conditional mean of Y given $X = x$ is linear in x . All parameters are population versions of the least squares estimates. In other words, the least squares estimators are the method of moments estimators of the parameters in the bivariate normal model.

1.6 Expectation and Variance of the parameter estimates

To be covered in tutorial.

1.7 Matrix notation

The model is $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$. In matrix form, $\underline{y} = X\beta + \underline{\epsilon}$:

$$\underline{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}_{n \times 1}, \quad X = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}_{n \times 2}, \quad \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \end{pmatrix}_{2 \times 1}, \quad \underline{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{pmatrix}_{n \times 1}$$

The least squares estimates are found by solving the normal equations, which are derived by minimizing the sum of squared errors.

$$\begin{aligned}\sum (y_i - \beta_0 - \beta_1 x_i) &= 0 \implies n\beta_0 + (\sum x_i)\beta_1 = \sum y_i \\ \sum x_i(y_i - \beta_0 - \beta_1 x_i) &= 0 \implies (\sum x_i)\beta_0 + (\sum x_i^2)\beta_1 = \sum x_i y_i\end{aligned}$$

In matrix form, this is written as:

$$X^T X \beta = X^T \underline{y}$$

The least squares estimates for β are given by:

$$\hat{\beta} = (X^T X)^{-1} X^T \underline{y}$$

The residuals are $\hat{\epsilon} = \underline{y} - \hat{\underline{y}} = \underline{y} - X\hat{\beta} = \underline{y} - X(X^T X)^{-1} X^T \underline{y}$

$$SSRes = \hat{\epsilon}^T \hat{\epsilon} = \underline{y}^T \underline{y} - \underline{y}^T X (X^T X)^{-1} X^T \underline{y}$$

$$SST = \underline{y}^T \underline{y} - \frac{1}{n^2} \underline{y}^T \underline{1} \underline{1}^T \underline{y}$$

where $\underline{1}$ is the vector of ones.